

Metals and Non-Metals

Complete Sequential Study Notes, Reactions & Metallurgical Concepts

1. Introduction & Physical Properties

A. Physical Properties of Metals

Metals are electropositive elements that form positive ions (cations) by losing valence electrons.

- **Hardness & Strength:** Generally hard with high tensile strength (e.g., Iron, Copper).
- **Physical State:** Solids at room temperature (*Exception:* Mercury (**Hg**) is a liquid metal at room temperature, known as *Quick Silver*).
- **Sonorous:** Produce a characteristic ringing sound when struck hard.
- **Conductivity:** Good conductors of heat and electricity due to free-moving electrons.
 - *Best Conductors:* Silver (**Ag**) and Copper (**Cu**).
 - *Poor Conductors:* Lead (**Pb**) and Mercury (**Hg**).
- **Malleability:** Can be beaten into thin sheets without breaking (e.g., Gold and Silver).
- **Ductility:** Can be drawn into long, thin wires (e.g., Gold is the most ductile metal).
- **Melting & Boiling Points:** High melting and boiling points (*Exceptions:* Caesium (**Cs**) and Gallium (**Ga**) have very low melting points and melt on palm).
- **Density:** High density (*Exceptions:* Alkali metals like **Li**, **Na**, **K**. **Osmium (Os)** has the highest density, **Lithium (Li)** has the lowest density).
- **Lustre:** Possess a shining metallic surface (metallic lustre).
- **Color:** Mostly silver-grey (*Exceptions:* Gold is yellow, Copper is reddish-brown).

B. Physical Properties of Non-Metals

Non-metals are electronegative elements that gain electrons to form negative ions (anions).

- **Physical State:** Exist in all three states of matter at room temperature:
 - *Solids:* Carbon, Sulphur, Phosphorus, Iodine.
 - *Liquid:* Bromine (**Br₂**) is the only liquid non-metal at room temperature.
 - *Gases:* Oxygen, Nitrogen, Hydrogen, Chlorine, Noble Gases.
- **Brittleness:** Solid non-metals are brittle (fragile) and break when hammered.
- **Ductility & Sonority:** Non-ductile and non-sonorous.
- **Conductivity:** Poor conductors of heat and electricity (*Exception:* Graphite, an allotrope of carbon, conducts electricity due to free electrons).
- **Lustre:** Dull appearance (*Exception:* Iodine is a lustrous non-metal).
- **Nature of Oxides:** Non-metallic oxides (e.g., **CO₂**, **NO₂**, **SO₂**) are **acidic** or neutral in nature.

Classification of Elements (Non-Metals & Metalloids):

- **Non-Metals (17 Elements):**
 - *Noble Gases:* Helium (**He**), Neon (**Ne**), Argon (**Ar**), Krypton (**Kr**), Xenon (**Xe**), Radon (**Rn**).
 - *Halogens:* Fluorine (**F₂**), Chlorine (**Cl₂**), Bromine (**Br₂**), Iodine (**I₂**), Astatine (**At**).
 - *Others:* Hydrogen (**H**), Carbon (**C**), Nitrogen (**N**), Oxygen (**O**), Phosphorus (**P**), Sulphur (**S**), Selenium (**Se**).
- **Metalloids (7 Elements):** Display properties intermediate between metals and non-metals: Boron (**B**), Silicon (**Si**), Germanium (**Ge**), Arsenic (**As**), Antimony (**Sb**), Tellurium (**Te**), Polonium (**Po**).

C. Summary of Exceptions in Physical Properties

Property	General Trend	Key Exceptions
Physical State	Metals are solid; Non-metals solid/gas	Mercury (Hg) is liquid metal. Bromine (Br₂) is liquid non-metal.
Hardness / Softness	Metals are hard and tough	Alkali metals (Na, K, Li) are soft and can be easily cut with a knife.
Melting Point	Metals have high melting points	Gallium (Ga) & Caesium (Cs) melt on palm. Diamond (non-metal) has high MP.
Lustre	Non-metals lack lustre	Iodine (I₂) is a lustrous non-metal.
Electrical Conduction	Non-metals do not conduct	Graphite (Carbon allotrope) conducts electricity.
Thermal Conduction	Metals conduct heat well	Lead (Pb) & Mercury (Hg) are poor conductors. Diamond is a great conductor.

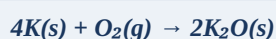
2. Chemical Properties of Metals

A. Reaction of Metals with Oxygen (Burnt in Air)

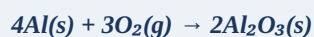
Metals react with oxygen to form **Basic Metal Oxides**.



- **Highly Reactive Metals (Na, K):** React vigorously with oxygen at room temperature and catch fire easily. They are kept immersed in **kerosene oil**.



- **Moderate Reactivity (Mg, Al, Zn, Pb):** React slowly and form a protective surface oxide layer that prevents further corrosion.



- **Iron Fillings:** Iron block does not burn on heating, but iron filings burn vigorously when sprinkled in a burner flame (attain ignition temperature faster).
- **Unreactive Metals (Ag, Pt, Au):** Do not react with oxygen even at high temperatures.

Basic & Amphoteric Oxides:

- **Basic Oxides:** Dissolve in water to form alkalis (turn red litmus blue):



- **Amphoteric Oxides:** Metal oxides that react with *both acids and bases* to produce salt and water (**Al₂O₃, ZnO, PbO, SnO**).

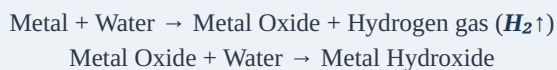
Aluminium Oxide (Al₂O₃):



Zinc Oxide (ZnO):



B. Reaction of Metals with Water or Steam

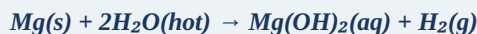


• Cold Water Reactions (K, Na, Ca):

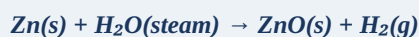


Floating Phenomenon: Calcium (Ca) and Magnesium (Mg) start **floating** on water because the bubbles of H_2 gas formed stick to their metal surfaces.

• Hot Water Reactions (Mg):

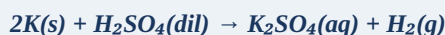


• Steam Reactions (Al, Zn, Fe): React only with steam to form metal oxides:



• No Reaction with Water: Lead (Pb), Copper (Cu), Silver (Ag), Gold (Au).

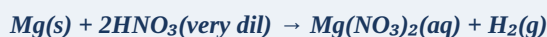
C. Reaction of Metals with Dilute Acids



Special Case - Dilute Nitric Acid (HNO_3):

Hydrogen gas (H_2) is **NOT evolved** when metals react with dilute nitric acid (HNO_3) because HNO_3 is a strong oxidizing agent; it oxidizes H_2 to water (H_2O) and itself gets reduced to nitrogen oxides (NO_2 , NO , N_2O).

Exception: Magnesium (Mg) and Manganese (Mn) react with very dilute HNO_3 (1-2%) to liberate H_2 gas:



Aqua-Regia (Royal Water):

A freshly prepared mixture of concentrated Nitric acid (HNO_3) and concentrated Hydrochloric acid (HCl) in a **1 : 3 ratio**. It is a highly corrosive, fuming liquid capable of dissolving gold (Au) and platinum (Pt).

D. Reaction of Metals with Bases

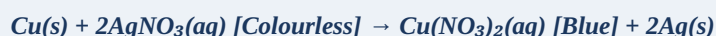
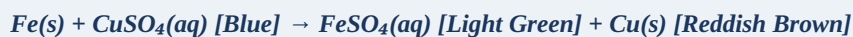
Strong bases react with specific active metals to liberate hydrogen gas:



E. Reaction of Metals with Salt Solutions (Displacement Reaction)



A more reactive metal displaces a less reactive metal from its aqueous salt solution.



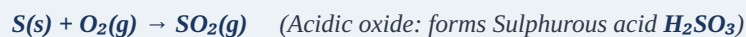
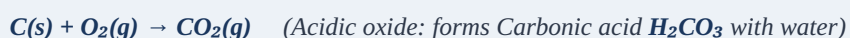
3. The Reactivity Series of Metals

Arrangement of metals in order of decreasing chemical reactivity.

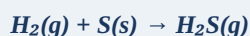
Symbol	Element	Reactivity Category	Chemical Behavior & Extraction Method
K	Potassium	HIGHLY REACTIVE METALS	React with cold water. Extracted by Electrolysis of molten ore.
Ba	Barium		
Ca	Calcium		
Na	Sodium		
Mg	Magnesium	MODERATELY REACTIVE METALS	Reacts with hot water.
Al	Aluminium		React with steam.
Zn	Zinc		Reacts with steam; reacts with acids.
Fe	Iron		
Ni	Nickel		React with dilute acids. Extracted by Carbon Reduction / Thermite Process.
Sn	Tin		
Pb	Lead		
H	Hydrogen	[Non-Metal Reference]	Reference standard for reactivity.
Cu	Copper	LOW REACTIVITY METALS	Do not react with water or dil. acids. Extracted by Self-reduction / Heating alone.
Hg	Mercury		
Ag	Silver		Found in Native / Free State in nature.
Au	Gold		
Pt	Platinum		

4. Chemical Properties of Non-Metals

- **Reaction with Oxygen:** Forms acidic or neutral non-metallic oxides.



- **Reaction with Water:** Non-metals **do not react with water** or steam because they cannot donate electrons to reduce H^+ ions into hydrogen gas.
- **Reaction with Dilute Acids:** Non-metals do not displace hydrogen from dilute acids.
- **Reaction with Hydrogen:** Combine with hydrogen to form covalent hydrides.



5. How Metals and Non-Metals React (Ionic Bonding)

Metals lose valence electrons to form cations, while non-metals gain electrons to form anions. The electrostatic force holding them together forms an **Ionic / Electrovalent Bond**.

A. Electronic Configurations & Ion Formation

• Group 1 (Alkali Metals):

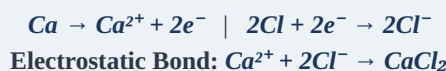
- Lithium (**Li**: 2, 1) → **Li⁺** (2) + **e⁻**
- Sodium (**Na**: 2, 8, 1) → **Na⁺** (2, 8) + **e⁻**
- Potassium (**K**: 2, 8, 8, 1) → **K⁺** (2, 8, 8) + **e⁻**
- Rubidium (**Rb**: 2, 8, 18, 8, 1)

• Group 2 (Alkaline Earth Metals):

- Beryllium (**Be**: 2, 2)
- Magnesium (**Mg**: 2, 8, 2) → **Mg²⁺** (2, 8) + **2e⁻**
- Calcium (**Ca**: 2, 8, 8, 2) → **Ca²⁺** (2, 8, 8) + **2e⁻**
- Strontium (**Sr**: 2, 8, 18, 8, 2)

B. Electron Transfer in Formation of Calcium Chloride (CaCl₂)

Calcium (**Ca**: 2, 8, 8, 2) loses 2 valence electrons to achieve Argon octet (2, 8, 8) and forms **Ca²⁺**. Two Chlorine atoms (**Cl**: 2, 8, 7) each accept 1 electron to form two **Cl⁻** ions (2, 8, 8).



C. Properties of Ionic Compounds

1. **Physical State:** Hard, brittle, crystalline solids owing to strong inter-ionic electrostatic forces.
2. **High Melting and Boiling Points:** Requires significant thermal energy to break the strong ionic lattice.
3. **Solubility:** Soluble in polar solvents like water; insoluble in non-polar solvents like petrol or kerosene.
4. **Electrical Conductivity:**
 - **Solid State:** Does **NOT** conduct electricity (ions are held tightly in fixed lattice positions).
 - **Molten / Aqueous State:** **Good conductor** (ions become free to move and carry electric charge).

6. Occurrence and Extraction of Metals (Metallurgy)

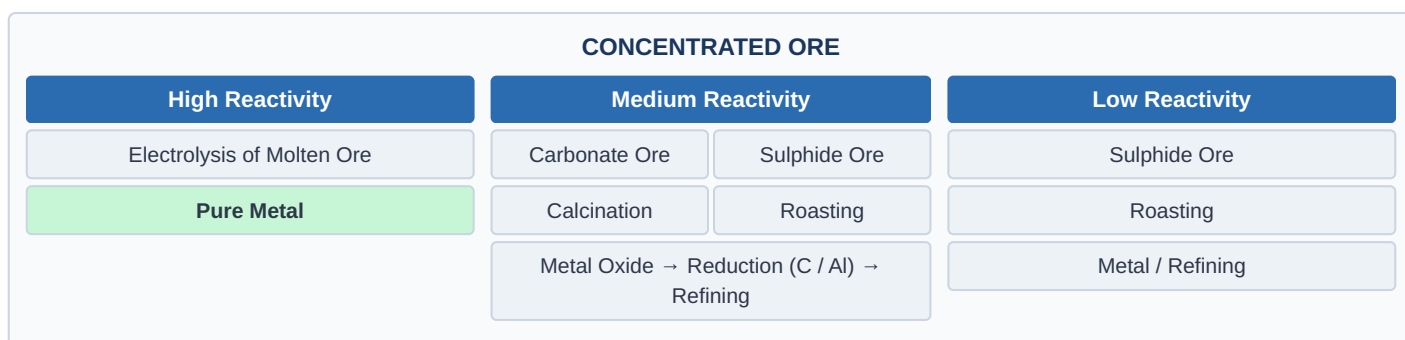
A. Ores and Minerals

- **Minerals:** Natural elements or inorganic compounds occurring in the earth's crust.
- **Ores:** Minerals from which a metal can be extracted commercially and **profitably**.
- **Gangue:** Soil, sand, clay, and rocky impurities present in mined ores.

Common Ores of Important Metals:

- **Aluminium (Al):** Bauxite (**Al₂O₃·2H₂O**)
- **Iron (Fe):** Haematite (**Fe₂O₃**), Magnetite (**Fe₃O₄**)
- **Copper (Cu):** Copper Glance (**Cu₂S**), Copper Pyrites (**CuFeS₂**)
- **Zinc (Zn):** Zinc Blende (**ZnS**), Calamine (**ZnCO₃**)
- **Lead (Pb):** Galena (**PbS**)
- **Mercury (Hg):** Cinnabar (**HgS**)

B. Metallurgical Flowchart



C. Comparison: Roasting vs. Calcination

Parameter	Roasting	Calcination
Ore Application	Mainly for Sulphide ores .	Mainly for Carbonate and Hydrated ores .
Presence of Air	Heated strongly in excess supply of air/oxygen .	Heated strongly in limited supply or absence of air .
Gas Released	Sulphur dioxide (SO_2) gas is liberated.	Carbon dioxide (CO_2) gas is liberated.
Chemical Equations	$2ZnS(s) + 3O_2(g) + heat \rightarrow 2ZnO(s) + 2SO_2(g)$	$ZnCO_3(s) + heat \rightarrow ZnO(s) + CO_2(g)$ $CaCO_3(s) + heat \rightarrow CaO(s) + CO_2(g)$

D. Extraction Techniques by Reactivity Class

1. Metals Low in Reactivity Series (Thermal / Self-Reduction)

- Mercury from Cinnabar (HgS):



- Copper from Copper Glance (Cu_2S):



2. Metals in Middle of Reactivity Series

- Smelting (Carbon / Coke Reduction):



- Thermite Reaction (Goldschmidt Process):

Exothermic reaction using Aluminium powder to reduce metal oxides:



Application: Thermite welding for railway tracks and machinery cracks.

3. Metals High in Reactivity Series (Electrolytic Reduction)

- Down's Process (Electrolysis of Molten NaCl):

- At Cathode (Reduction): $Na^+(molten) + e^- \rightarrow Na(s)$ (Metal deposited)
- At Anode (Oxidation): $2Cl^-(molten) \rightarrow Cl_2(g) + 2e^-$ (Chlorine gas evolved)

- Hall-Héroult Process (Electrolysis of Alumina Al_2O_3 in Molten Cryolite Na_3AlF_6):

- At Cathode (Reduction): $2Al^{3+} + 6e^- \rightarrow 2Al(s)$
- At Anode (Oxidation): $6O^{2-} \rightarrow 3O_2(g) + 12e^-$

E. Electrolytic Refining of Copper

- Anode:** Impure Copper block.
- Cathode:** Thin sheet of Pure Copper.
- Electrolyte:** Acidified Copper Sulphate ($CuSO_4$) solution.
- Reactions:**



- **Anode Mud:** Insoluble impurities (e.g., Gold, Silver) settling under the anode.

7. Corrosion & Alloys

A. Corrosion Processes

- **1. Rusting of Iron:**



- **2. Corrosion of Copper (Patina):**



- **3. Tarnishing of Silver:**



B. Corrosion Prevention Methods

1. **Barrier Methods:** Painting, oiling, greasing, or varnishing metal surfaces.
2. **Galvanization:** Coating iron or steel articles with a protective layer of **molten Zinc**.
3. **Electroplating:** Depositing a thin layer of superior metal (e.g., Chromium, Silver, Gold).
4. **Sacrificial Protection:** Attaching more active metals like Magnesium or Zinc, which corrode preferentially.
5. **Alloying:** Mixing metals with other elements to enhance chemical stability.

C. Alloys and Their Applications

An alloy is a **homogeneous mixture** of two or more metals, or a metal and a non-metal.

Alloy	Composition	Key Properties & Applications
Steel	Iron + Carbon ($\text{Fe} + \text{C} \sim 0.05\%$)	Hard, tough, high tensile strength.
Stainless Steel	Iron + Nickel + Chromium ($\text{Fe} + \text{Ni} [18\%] + \text{Cr} [18\%] + \text{C}$)	Rust resistant; surgical tools, cookware.
Brass	Copper + Zinc ($\text{Cu} \sim 70\% + \text{Zn} \sim 30\%$)	Malleable, acoustic properties; instruments, hardware.
Bronze	Copper + Tin ($\text{Cu} \sim 90\% + \text{Sn} \sim 10\%$)	Corrosion resistant; statues, coins, medals.
Solder	Lead + Tin ($\text{Pb} \sim 50\% + \text{Sn} \sim 50\%$)	Low melting point; welding electrical wires.
Amalgam	Mercury + Metal ($\text{Hg} + \text{Na} / \text{Ag}$)	Dental fillings and specialized chemical processes.